

"Grade XII: Chemistry"

1st Term Solution

1st-Term Chemistry Paper Solution 2081

Group 'C' Long Question

Long Questions

By Pankaj Purbey

Q.N.9. (a) Ans:

Role of HCl in the qualitative analysis of Group II metal sulphide:

Both Group II & Group III B metal ions precipitate out in the form of metal sulphide. However, solubility product of Group II metal sulphide is lesser than that of Group III B metal sulphide. When HCl is added along with H_2S , concentration of sulphide ion in solution decreases due to common ion effect. This decrease in sulphide ion concentration causes decrease in ionic product of metal sulphide. This decreased ionic product exceeds solubility product of only Group II metal sulphide but not of Group III B metal sulphide. So, precipitation of only Group II metal sulphide occurs leaving Group III B metal ions in the solution.

If HCl is not added, then the concentration of sulphide ion in solution increases. This causes increase in ionic product of metal sulphide. This increased ionic product exceeds solubility product

of both Group II and Group III B metal sulphide. So, precipitation of both Group II and Group III B metal sulphide occurs.

Hence, use of HCl is in the selective precipitation of Group II metal sulphide from the mixture of Group II and Group III B metal ions.

E.g: $\text{H}_2\text{S} \rightleftharpoons 2\text{H}^+ + \text{S}^{2-}$
 $\text{HCl} \rightleftharpoons \text{H}^+ + \text{Cl}^-$

9. (b) Solution:

For AgNO_3 ,

Before Mixing

$$V_1 = 200 \text{ ml}$$

$$M_1 = 1.3 \times 10^{-3} \text{ M}$$

After Mixing

$$V_2 = 300 \text{ ml}$$

$$M_2 = ?$$

We have,

$$M_1 V_1 = M_2 V_2$$

$$\therefore M_2 = \frac{M_1 V_1}{V_2} = \frac{1.3 \times 10^{-3} \times 200}{300} = 8.67 \times 10^{-4} \text{ M}$$

$$\text{Concentration of } \text{AgNO}_3 = 8.67 \times 10^{-4} \text{ M}$$



$$8.67 \times 10^{-4} \text{ M}$$

$$8.67 \times 10^{-4} \text{ M}$$

$$\therefore [\text{Ag}^+] = 8.67 \times 10^{-4} \text{ M}$$

For Na_2S ,

Before Mixing

$$V_1 = 100 \text{ ml}$$

$$M_1 = 4.5 \times 10^{-5} \text{ M}$$

After Mixing

$$V_2 = 300 \text{ ml}$$

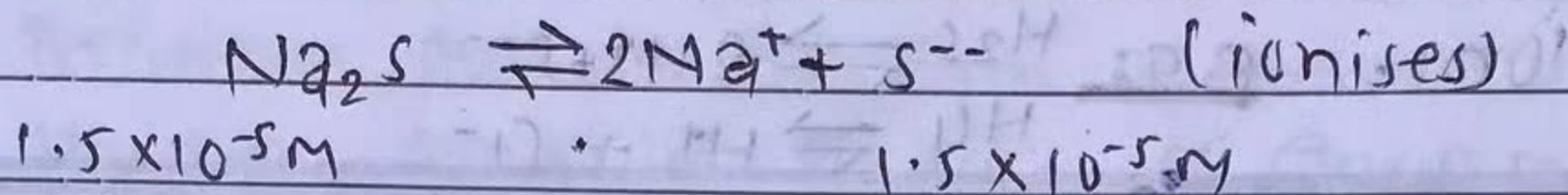
$$M_2 = ?$$

We have,

$$M_1 V_1 = M_2 V_2$$

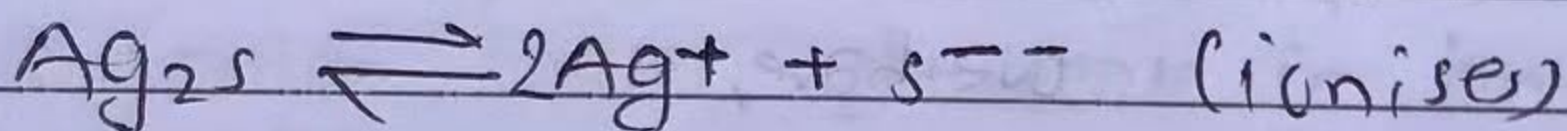
$$\therefore M_2 = \frac{M_1 V_1}{V_2} = \frac{4.5 \times 10^{-5} \times 100}{300} = 1.5 \times 10^{-5} M$$

Concentration of $Na_2S = 1.5 \times 10^{-5} M$



$$\therefore [S^{--}] = 1.5 \times 10^{-5} M$$

Given, K_{sp} of $Ag_2S = 1.6 \times 10^{-49}$



$$\begin{aligned} K_{sp} &= [Ag^+]^2 [S^{--}] \\ &= (8.67 \times 10^{-4})^2 \times (1.5 \times 10^{-5}) \\ &= 1.12 \times 10^{-11} \end{aligned}$$

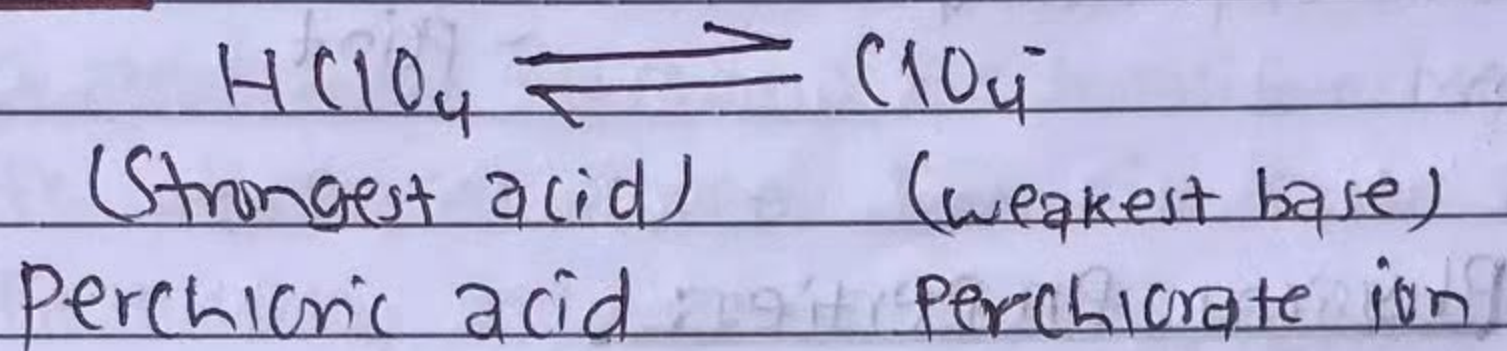
Since, $K_{sp} > K_{ip}$ $\therefore$ So, precipitation occurs.

9. (U) Ans: Bronsted - Lowry concept define acid & base on the basis of tendency of a substance to donate or accept proton. But conjugate acid-base pair is the pair of acid and base which are

interconvertible to each other either by donating or accepting proton.

From the concept of conjugate acid-base pair, relative strength of acid and base can be determined. If Bronsted acid or base is strong, its conjugate acid or base will be weak and vice-versa.

for eg:



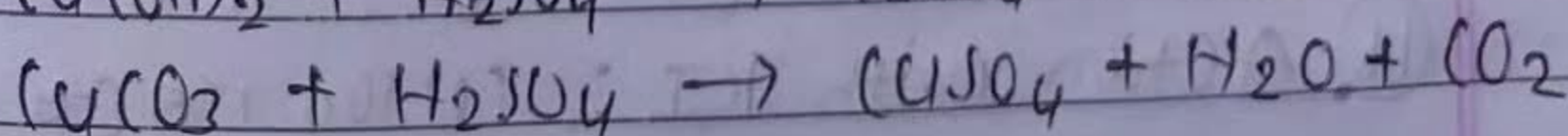
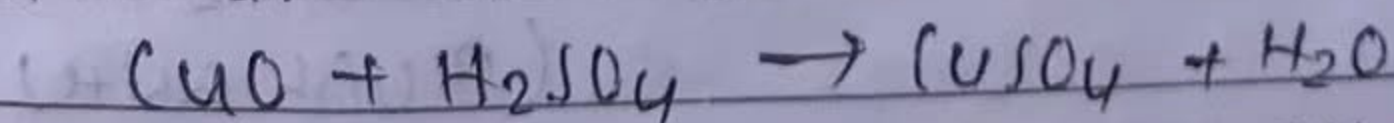
Q. N. 10. (a) Ans:

"Chemistry of Blue vitriol"

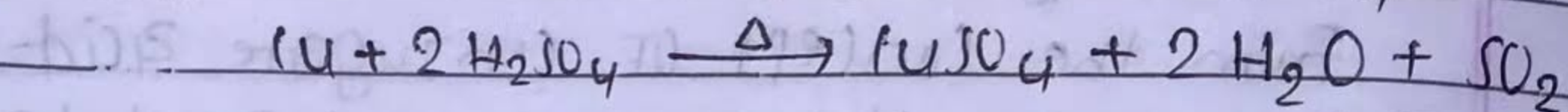
Copper sulphate pentahydrate is called blue vitriol. Its molecular formula is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

Preparation:

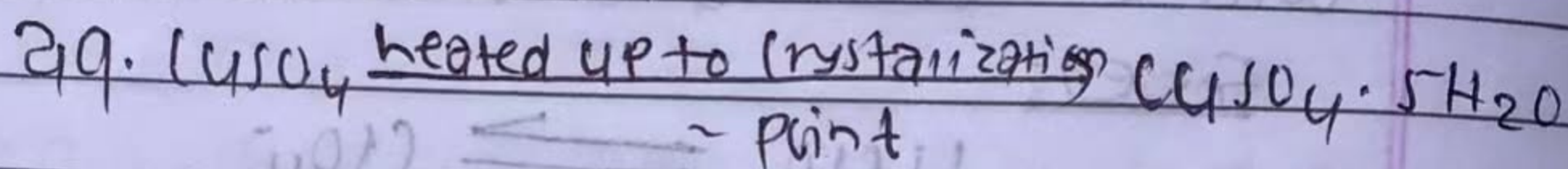
- ① It is prepared by reacting its oxide, hydroxide or carbonate with dil. H_2SO_4 in lab.



(II) It can be also prepared by action of copper with hot and conc. H_2SO_4 .



(III) It is evaporated up to crystallization point followed by cooling to obtain crystal of blue vitriol.



Physical Properties:

- (i) It is blue crystalline solid.
- (ii) It is readily soluble in water.
- (iii) Its aqueous solution is highly acidic in nature.
- (iv) Anhydrous $CuSO_4$ is white.

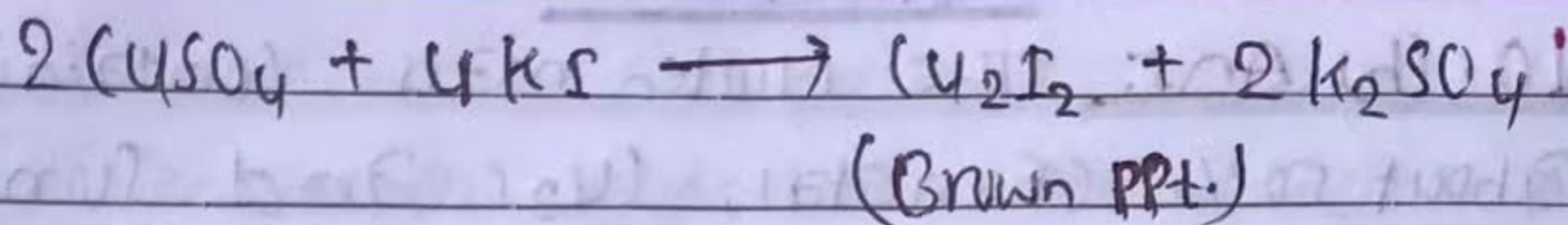
Chemical Properties:

- (i) Action with alkalis: with $NaOH$ solution, it gives bluish-white ppt of cupric hydroxide.

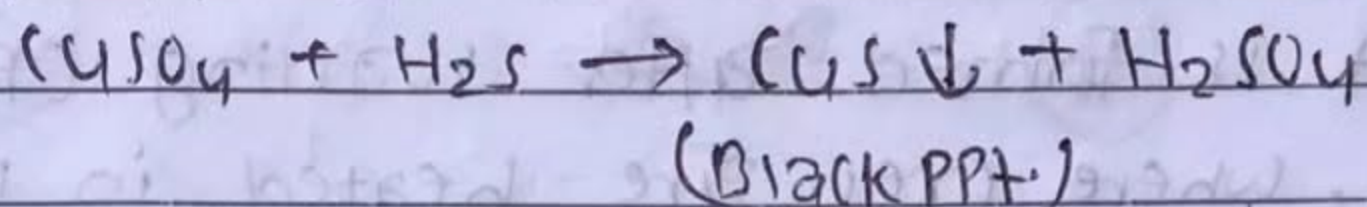
$$CuSO_4 + 2NaOH \rightarrow Cu(OH)_2 \downarrow + Na_2SO_4$$

(Bluish white)

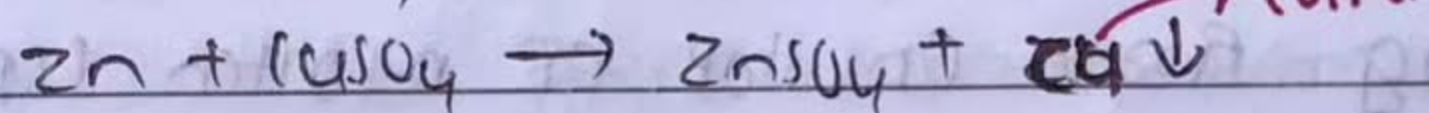
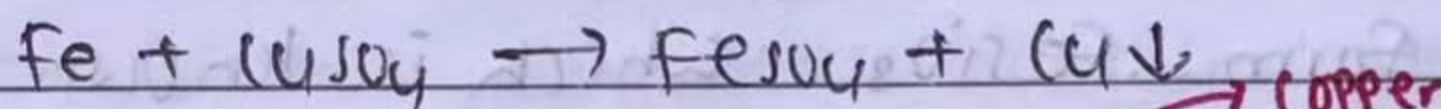
- (ii) Action with KI: $CuSO_4$ reacts with KI to form dark brown ppt of Cu_2I_2 is formed.



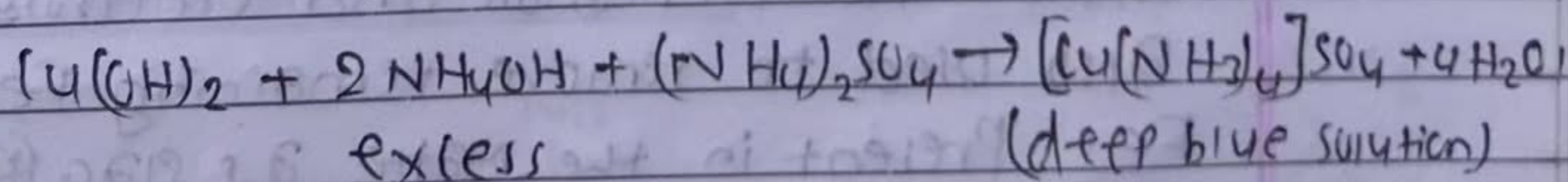
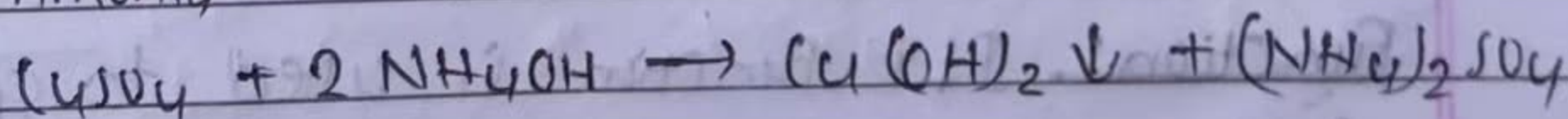
III Action with H_2S : with H_2S gas, it gives black ppt of cupric sulphide.



IV Displacement reaction: Electropositive metals like Zn, Fe displaces copper from its salt solution.



V Action with ammonia: It gives deep blue solution of tetramminecopper (II) sulphate when treated with an aqueous solution of excess ammonia.



Uses:

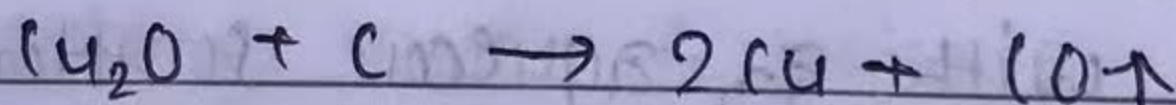
- (i) In medicine as an antiseptic.
- (ii) It is used in electroplating.
- (iii) It is used as fungicide and germicide.
- (iv) It is used in dyes and pigment industry.

Definition:

10. (b) Ans: The Molten mass containing about 50% of metal, Cu_2S and some trace of FeS is called matte which forms lower layer in the blast furnace.

Formation:

It is formed after roasting of copper ores, where ores are heated in presence of oxygen. After roasting, roasted ore is mixed with silica (SiO_2) and introduced into blast furnace. Silica reacts to FeO to form FeSiO_3 which is known as slag. Slag floats on top of molten copper matte allowing for easier separation.



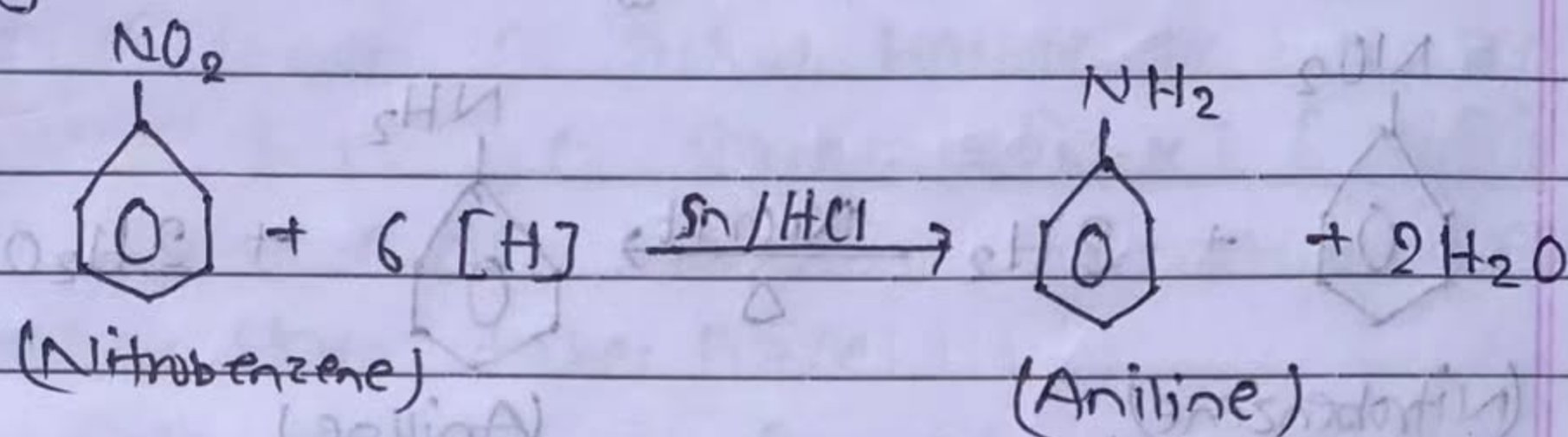
Function: It's function is used to give pure copper after purification and electrolysis. It is also used to remove remaining FeO and FeS present in the matte as slag (FeSiO_3).

Q. N. 11. (a) Ans:

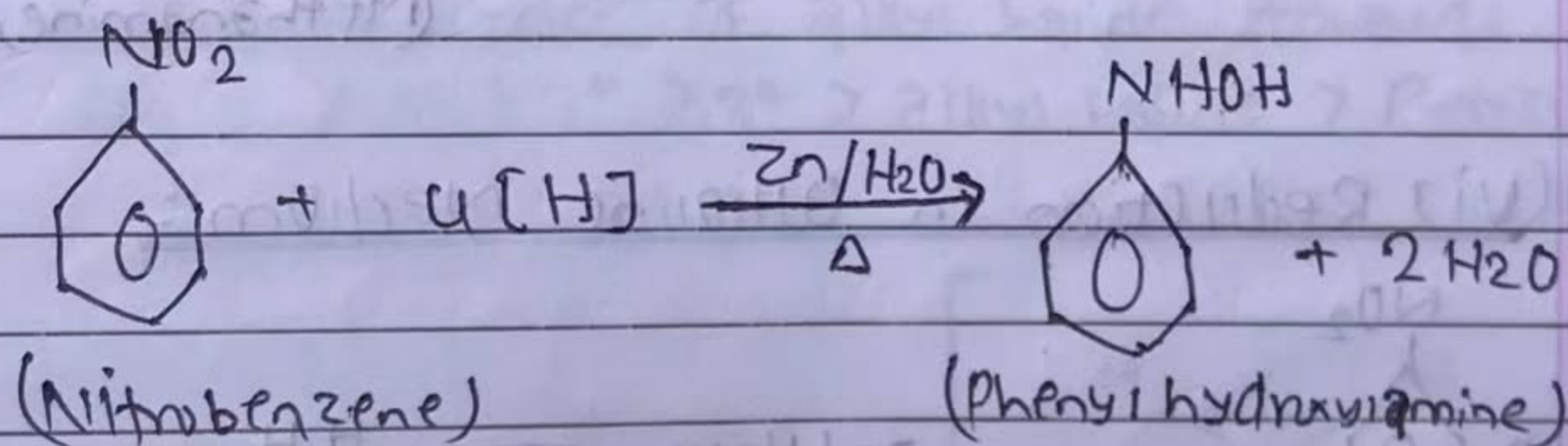
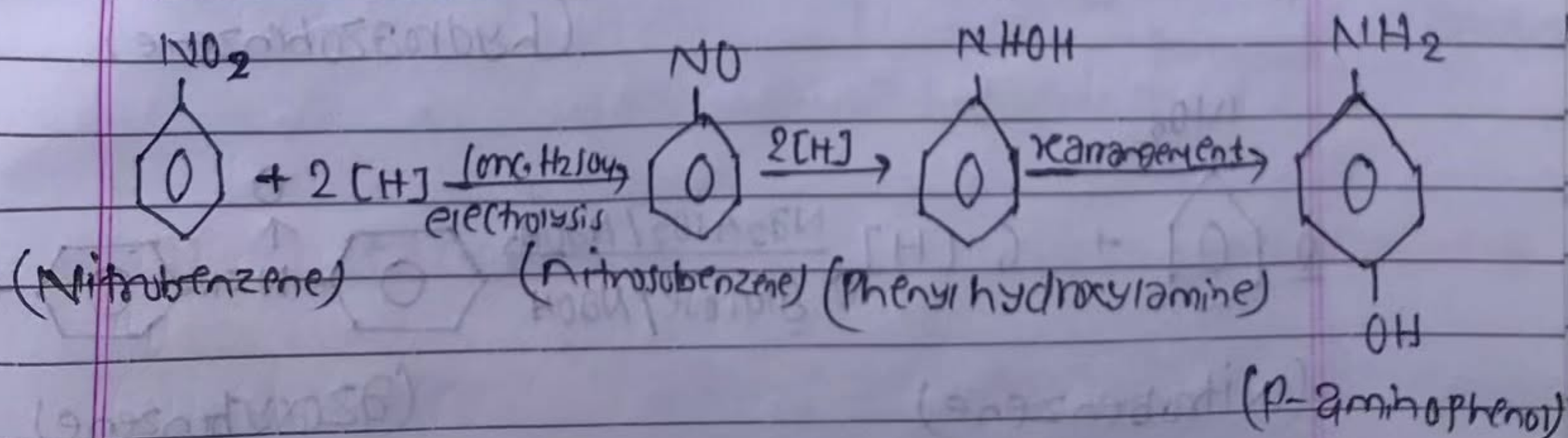
'Reduction of nitrobenzene in various medium':

(i) Reduction in Acidic medium:

Nitrobenzene on reduction in acidic medium gives aniline.

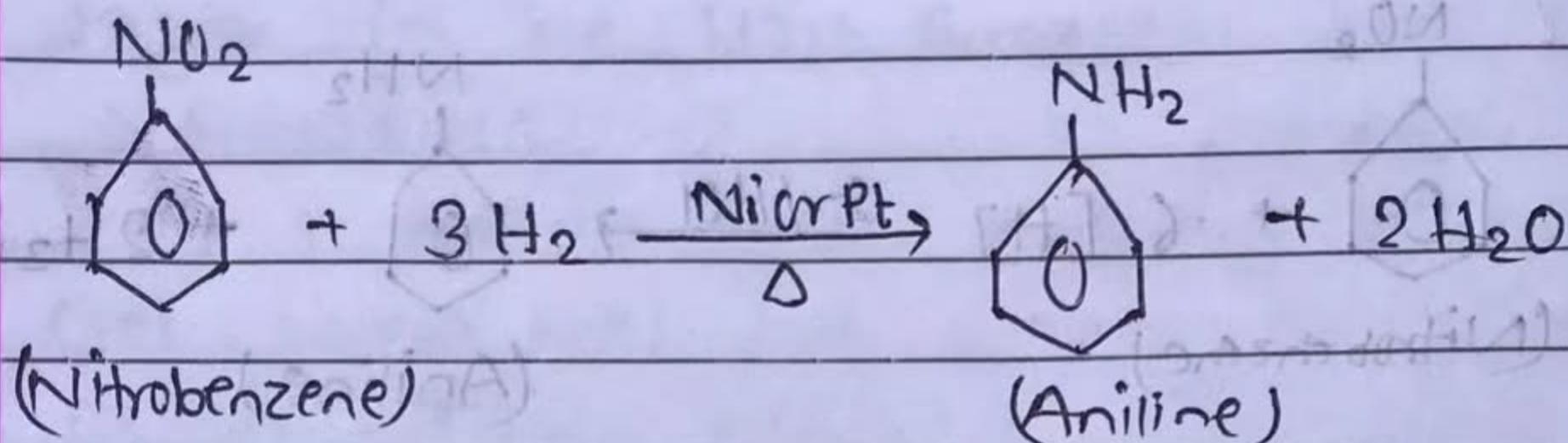
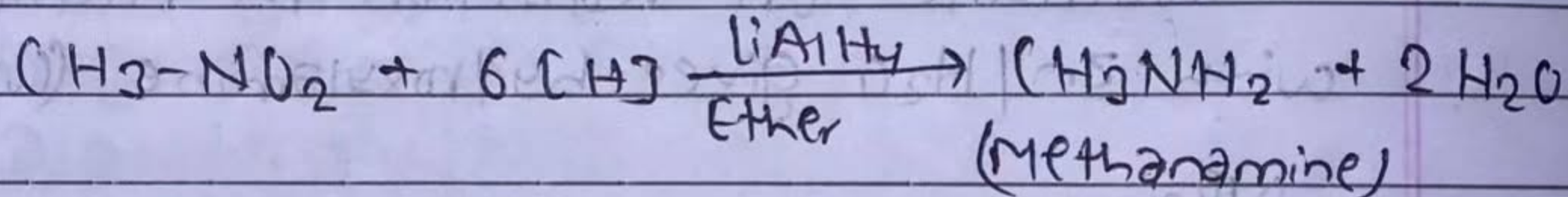
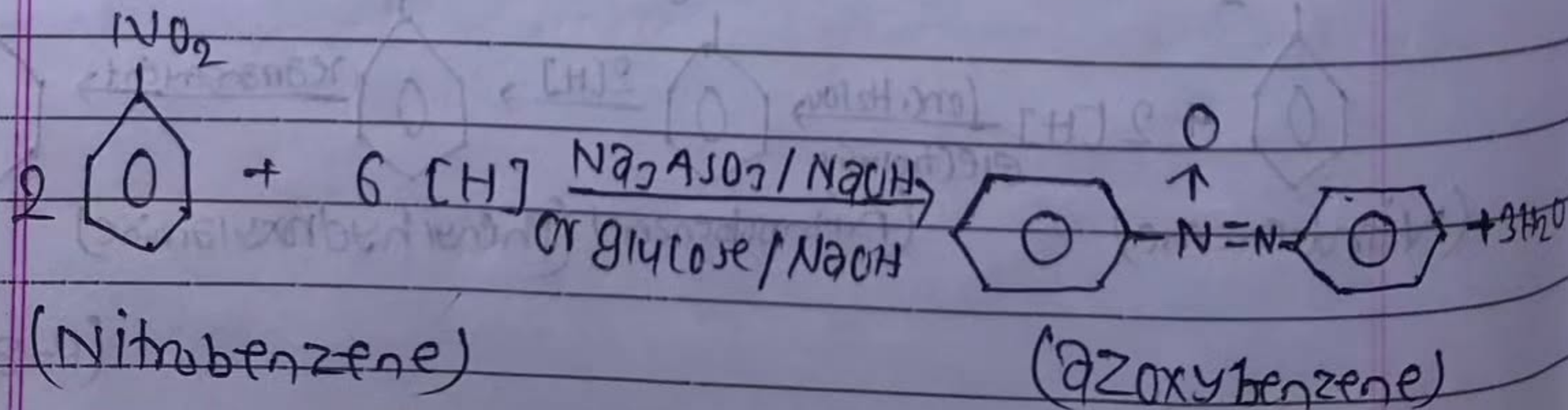
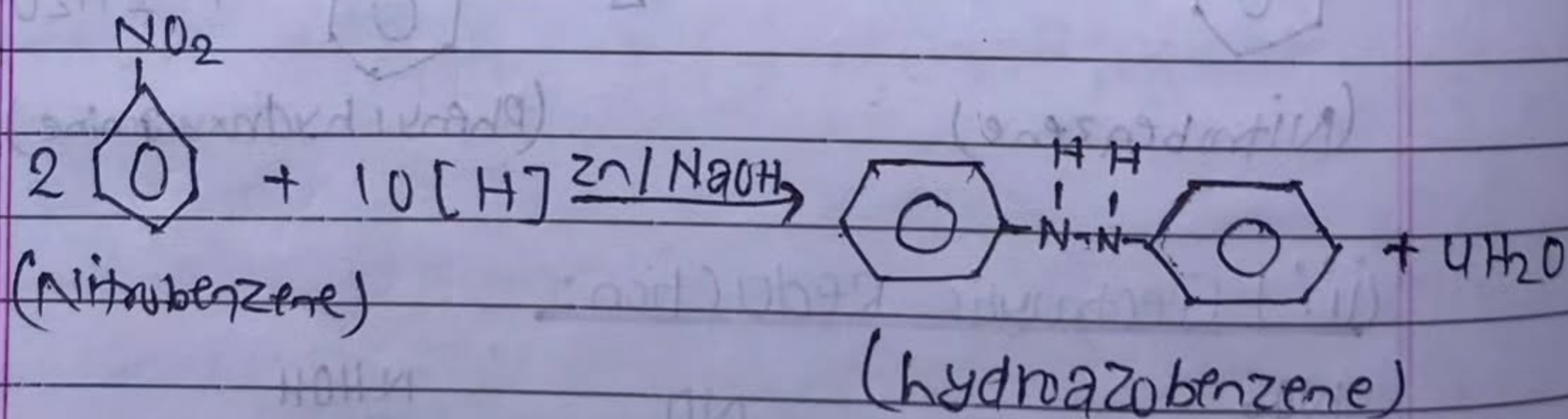
(ii) Reduction in Neutral medium:

Nitrobenzene on reduction in neutral medium with $\text{Zn/H}_2\text{O}$ gives hydroxylamine (unresolving).

(iii) Electrolytic Reduction:

(iv) Catalytic reduction:

Nitrobenzene on catalytic reduction with hydrogen in presence of Ni, Pt gives aniline.

(v) Reduction with LiAlH_4 :(vi) Reduction in alkaline medium:

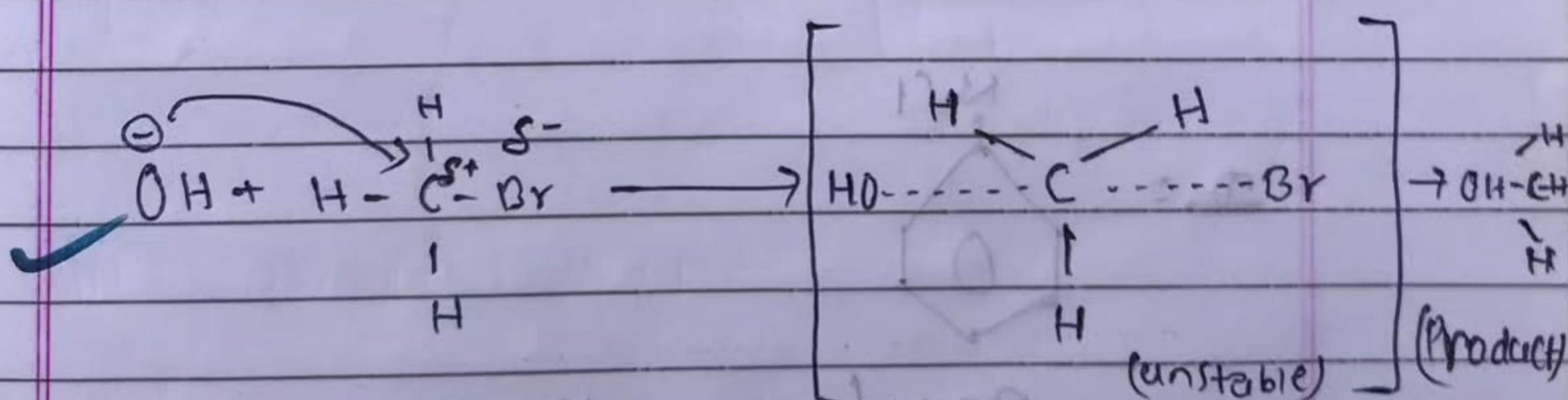
11. (b) Ans: 'S_N2 Reaction':

This reaction proceeds in one step. It is bimolecular reaction. Rate of reaction depends on both concentration of alkyl halide as well as nucleophile. i.e. $\text{Rate} = k[\text{R-X}][\text{Nu}^-]$. It is second order reaction. Formation of transition state takes place.

S_N2 reaction is favoured by non-polar solvent. Nucleophile attacks take place from opposite side of leaving group. So, inverted product is formed and this inversion is called Walden inversion.

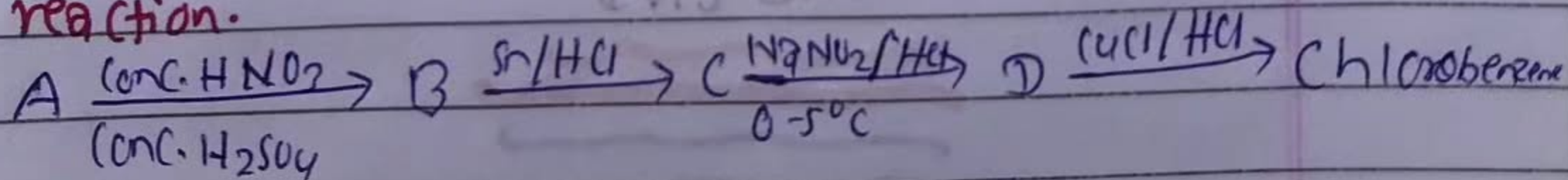
Reactivity order of alkyl halide towards S_N2 rxn:

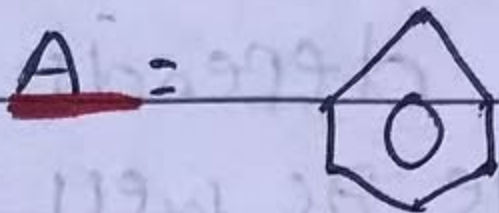
$(\text{H}_3\text{-X} > 1^\circ > 2^\circ > 3^\circ > \text{alkyl halide} > \text{Benzyl halide})$



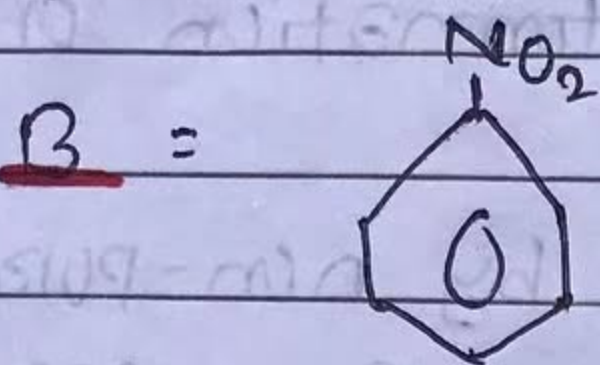
It is due to steric hindrance.

11. (c) Identify A, B, C and D in the following reaction.

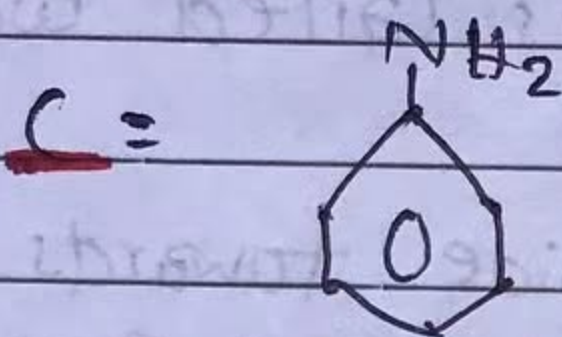


Ans:

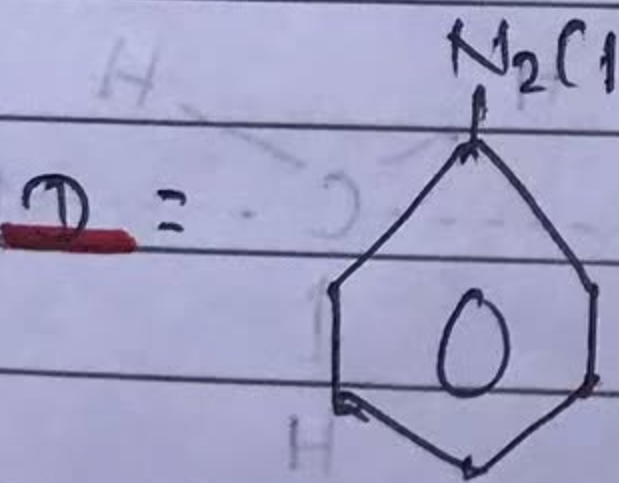
(Benzene)



(Nitrobenzene)



(Aniline)



BDC (Benzene diazonium chloride)

"The End"

KMC Note

Symbol no.:

Invigilator's Sign:

Re-write the correct option of each question in your answer sheet.

GROUP-A (MCQ) [11×1 = 11]

- The volume of water that must be added to 50 ml of 49 gm/lit sulphuric acid solution to make it exactly semi normal is
a) 100 ml ☒ b) 50 ml c) 150 ml d) 200 ml
- The equivalent mass of H_3PO_4 in the reaction,
 $H_3PO_4 + Ca(OH)_2 \rightarrow CaHPO_4 + 2H_2O$ is
a) 98 b) 32.6 ☒ c) 49 d) 37
- A buffer solution of pH=9 can be prepared by mixing
a) CH_3COONa & CH_3COOH b) $NaCl$ & $NaOH$
☒ c) NH_4Cl & NH_4OH d) KH_2PO_4 & H_3PO_4
- The pH of a neutral solution at $50^\circ C$ is ($pK_w=13.26$)
☒ a) 6.63 b) 6.03 c) 7.13 d) 7.0
- The most stable oxidation state of chromium is
a) +5 ☒ b) +3 c) +2 d) +4
- Formula of Schweitzer's reagent is
☒ a) $[Cu(NH_3)_4]SO_4 \cdot 4H_2O$ b) $[Cu(NH_3)_4]SO_4 \cdot 5H_2O$
c) $[Cu(NH_3)_4]SO_4 \cdot H_2O$ d) $[Cu(NH_3)_4]SO_4$
- Which of the following is used to produce lithopone?
a) $ZnCO_3$ b) ZnO c) $ZnCl_2$ ☒ d) $ZnSO_4$
- Lucas reagent is
☒ a) anhydrous $ZnCl_2$ + Conc. HCl b) conc. H_2SO_4
c) alc. KOH d) $Zn - Hg/HCl$
- Chloral is needed for the synthesis of
a) Benzal Chloride ☒ b) DDT c) Chlorobenzene d) Gammexane
- Acetic anhydride reacts with diethyl ether in the presence of anhydrous $AlCl_3$ to give
a) CH_3COOCH_3 b) $CH_3CH_2COOCH_3$
☒ c) $CH_3COOCH_2CH_3$ d) CH_3CH_2OH
- The electrophile in nitration is
a) nitrate ion b) nitrite ion c) cyanide ion ☒ d) nitronium ion

Class: XII 'Science'

Time: 25 Min.

Subject: Chemistry (3021)

Tick the best options among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

KMC Note

Symbol no.:

Invigilator's Sign:

Re-write the correct option of each question in your answer sheet

GROUP-A (MCQ) [11×1 = 11]

- Acetic anhydride reacts with diethyl ether in the presence of anhydrous $AlCl_3$ to give
 - CH_3COOCH_3
 - $CH_3CH_2COOCH_3$
 - ☒ $CH_3COOCH_2CH_3$
 - CH_3CH_2OH
- Lucas reagent is
 - ☒ anhydrous $ZnCl_2$ + Conc. HCl
 - conc. H_2SO_4
 - alc. KOH
 - $Zn - Hg/HCl$
- Chloral is needed for the synthesis of
 - Benzal Chloride
 - ☒ DDT
 - Chlorobenzene
 - Gammexane
- The electrophile in nitration is
 - nitrate ion
 - nitrite ion
 - cyanide ion
 - ☒ nitronium ion
- The pH of a neutral solution at $50^\circ C$ is ($pK_w = 13.26$)
 - ☒ 6.63
 - 6.03
 - 7.13
 - 7.0
- A buffer solution of pH=9 can be prepared by mixing
 - CH_3COONa & CH_3COOH
 - $NaCl$ & $NaOH$
 - ☒ NH_4Cl & NH_4OH
 - KH_2PO_4 & H_3PO_4
- The equivalent mass of H_3PO_4 in the reaction, $H_3PO_4 + Ca(OH)_2 \rightarrow CaHPO_4 + 2H_2O$ is
 - 98
 - 32.6
 - ☒ 49
 - 37
- The volume of water that must be added to 50 ml of 49 gm/lit sulphuric acid solution to make it exactly semi normal is
 - 100 ml
 - ☒ 50 ml
 - 150 ml
 - 200 ml
- Formula of Schweitzer's reagent is
 - ☒ $[Cu(NH_3)_4]SO_4 \cdot 4H_2O$
 - $[Cu(NH_3)_4]SO_4 \cdot 5H_2O$
 - $[Cu(NH_3)_4]SO_4 \cdot H_2O$
 - $[Cu(NH_3)_4]SO_4$
- The most stable oxidation state of chromium is
 - +5
 - ☒ +3
 - +2
 - +4
- Which of the following is used to produce lithopone?
 - $ZnCO_3$
 - ZnO
 - $ZnCl_2$
 - ☒ $ZnSO_4$

KMC Note

Class: XII 'Science'

F.M: 75

Time: 3hr.

Subject: Chemistry [3021]

P.M: 30

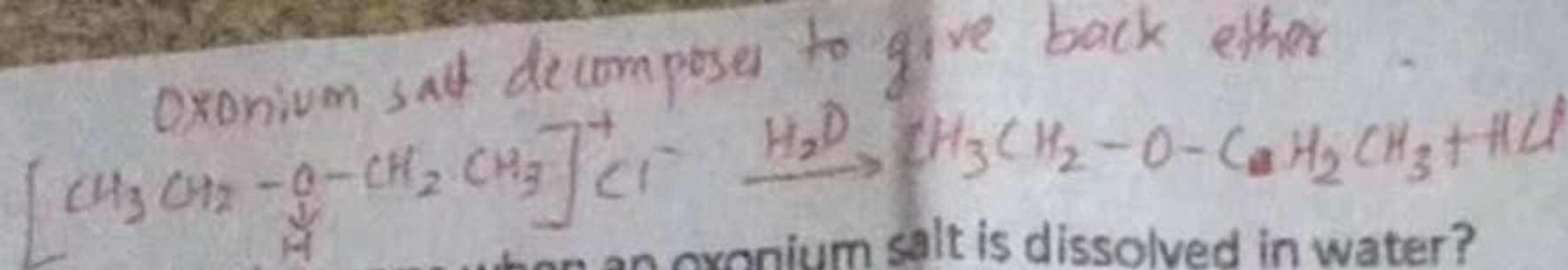
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP B [8×5=40]

Very short questions

1. a) State law of chemical equivalence and hence derive normality equation. [1+2]
 b) 7.875 gm hydrated oxalic acid crystals are dissolved in 250 ml water. 16 ml of this solution neutralizes 40 ml of 0.2N NaOH. Calculate the number of water of crystallization. 2 [2]
 $V_1 N_1 = V_2 N_2$ or $V_1 S_1 = V_2 S_2$
2. a) Derive an expression of Ostwald's dilution law for weak electrolyte and write its limitation. $\alpha = \sqrt{\frac{K_{eq}}{C}}$ or $\alpha = \sqrt{K_{eq} \times V}$ Applicable only for weak electrolyte [2+1]
 b) The pH of KOH is 8. If it is diluted 1000 times, calculate the pH of diluted base. [2]
 $pOH = 6.995$ $pH = 7.004$
3. a) Discuss the effect of temperature and catalyst in reaction rate. [1+2]
 b) Differentiate between average rate of reaction and instantaneous rate of reaction. At least 3 points. $\rightarrow$ certain interval of time $\rightarrow$ particular instant of time [2]
 $\rightarrow$ Energy profile diagram for catalyzed & uncatalyzed rxn is reqd.
4. a) Write short note on 'Extraction of Zinc from its chief ore.' [4]
 b) What happens when zinc is exposed to moist air? $Zn(O_2 \cdot Zn(OH)_2)$ is formed. Basic zinc carbonate [1]
KMC Note
5. List out the main general characteristics of Transition metals with relevant examples. [5]
6. a) Starting from trichloromethane, how would you prepare
 i) Ethyne ii) Chloropicrin iii) chloretone [1+1+1]
 b) How would you distinguish ethanol from propan-1-ol. [2]
7. a) Write the formula of various positional and functional isomers of alcohols having molecular formula C_3H_8O with IUPAC name. [2]
 b) How would you distinguish these isomers by Victor Meyer's method? [3]
8. a) Give reason
 i) It is dangerous to boil old sample of ether. $CH_3CH_2OCH_2CH_3 + O_2 \xrightarrow[\text{air}]{\text{Sunlight}} CH_3CH_2-O-O-CH_2CH_3$ peroxide of ether
 ii) Ether is stored in bottle containing iron wire. [1+1]
 $\rightarrow$ Iron has greater affinity towards oxygen than ether has.
 $\rightarrow$ So, iron combines with oxygen and formation of peroxide is controlled.



- b) What happens when an oxonium salt is dissolved in water? [1]
- c) Why chlorobenzene is less reactive as compared to benzene towards electrophilic substitution reaction? [2]

KMC Note

Resonating structure of chlorobenzene is reqd.

Group C [3x8=24]

Long Questions

9. a) What is the role of dil. HCl in the qualitative analysis of Group-II Metal sulphide? Explain. [3]

In the selective precipitation of Gr II Metal sulphide. Explanation reqd.

b) 200 ml of $1.3 \times 10^{-3} M AgNO_3$ is mixed with 100 ml of $4.5 \times 10^{-5} M Na_2S$. Will precipitation occur? (K_{sp} of $Ag_2S = 1.6 \times 10^{-49}$) Yes ppt occurs. [3]

c) Explain the concept of relative strength of acid and base with relevant examples on the basis of Bronsted-Lowry concept. [2]

(Strongest acid) $HClO_4 \rightleftharpoons ClO_4^-$ (Weakest base) Strong bronsted acid will form weak conjugate base & vice versa.

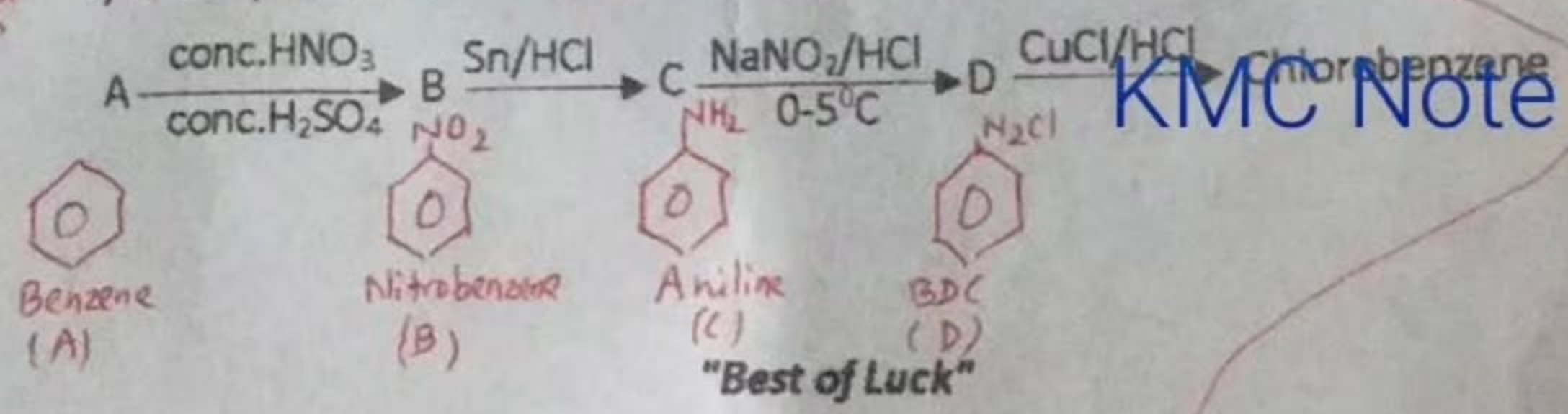
10. a) Write short note on "Chemistry of Blue Vitriol". [5]

b) What is 'matte'? How is it formed? What is its function? [3]

11. a) Write short note on "Reduction of nitrobenzene in various medium" [4]

b) Explain SN_2 reaction. (i) Rate $\propto [R-X][Nu^-]$ (ii) 2nd order (iii) Reactivity order: $CH_3X > 1^\circ > 2^\circ > 3^\circ$ alkyl halide [2]

c) Identify A, B, C and D in the following reaction. [2]



KMC Note

